

LESSON 2-9 • SESSION 1 • PRACTICE • FORM A

Finding Mean Absolute Deviation (MAD)

Learning target – I can find the mean absolute deviation of a data set and explain what it measures. · 6.SP.B.5c

REMEMBER

- **MAD** (mean absolute deviation) = the mean of the distances from the mean.
- Steps: find the **mean**, find each value's **distance** from it (never negative), then **average** the distances.
- If all values match, the MAD is 0. A **small** MAD means values sit close to the mean.

WORKED EXAMPLE

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

Step 1 Mean: $28 \div 4 = 7$.**Step 2** Distances from 7: 4, 0, 1, 3.**Step 3** Sum = 8, and $8 \div 4 = 2$.**Answer** MAD = 2 cm

WORDS mean absolute deviation (*desviación media absoluta*) · deviation (*desviación*) · absolute value (*valor absoluto*) · spread (*dispersión*)

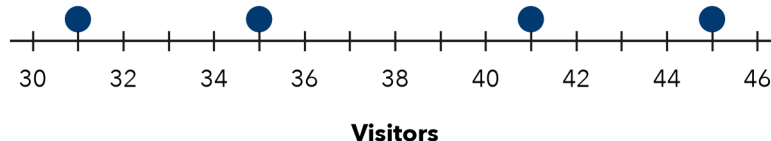
PRACTICE

show your work · label units

1. **Science Quiz.** The mean of 13, 15, 19, 23 and 25 is 19. Complete the table. Then find the MAD: _____

Value	13	15	19	23	25
Distance from 19					

2. **Gym Visits.** The dot plot shows the visitors to a gym on four evenings.



- a. Mean: _____ visitors
- b. MAD: _____ visitors
3. **Afternoon Temps.** Four afternoon temperatures were 53, 57, 59 and 63°F. Find the MAD.

Hint Find the mean first. Measure each distance from it.

Show your work.

Answer _____ °F

4. **All the Same.** Four values are 25, 25, 25, 25. Elise says the MAD cannot be found. Is she right?

Hint Find each value's distance from the mean.

Show your work.

MAD _____

5. **Reading a MAD.** Six delivery times have a mean of 45 minutes and a MAD of 3 minutes. What does the MAD tell a customer?

Show your work.

Meaning _____

- 6. Homework Pages.** Five students turned in 12, 18, 20, 22 and 28 pages. Find the mean and the MAD.

Hint Measure distances from the mean, not from 0.

Show your work.

Mean, MAD _____

- 7. Missing Distance.** Four values have mean 50 and MAD 5. Three distances from the mean are 6, 2 and 4. Find the fourth distance.

Hint $MAD \times 4$ gives the total of the distances.

Show your work.

Distance _____

- 8. Two Groups.** Free throws made – Group 1: 6, 8, 10, 12. Group 2: 4, 9, 10, 13. Both means are 9.

a. MAD of Group 1: _____

b. MAD of Group 2: _____

c. What does the MAD tell you that the mean does not? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Damaris's Work.** For 17, 19, 25, 27 (mean 22), Damaris added -5 , -3 , $+3$, $+5$ and reported a MAD of 0. Explain her mistake and find the MAD.

TEST PRACTICE

choose the best answer

- 10.** MAD stands for –

A middle and difference

B mean added data

C median absolute distance

D mean absolute deviation

- 11.** Distances from the mean are never –

A zero

B negative

C whole numbers

D decimals

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

0	2.5	12	8	4	25	20	5	3	10
I	E	Q	T	D	Z	A	E	V	X

#1	#2b	#3	#4	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____